Example 6.4

The following are the weights of 10 children admitted in a hospital on a particular day.

Find the mean deviation about mean, median and their coefficients of mean deviation.

7, 4, 10, 9, 15, 12, 7, 9, 9, 18

Solution:

$n = 10$; Mean: $\bar{x} = \frac{\sum x}{n} = \frac{100}{10} = 10$

Median: The arranged data is: 4, 7, 7, 9, 9, 9, 10, 12, 15, 18

Median $= \frac{9+9}{2} = \frac{18}{2} = 9$:

Marks (x)	**$\lvert D \rvert = \lvert x - \bar{x} \rvert$**	**$\lvert D_m \rvert = \lvert x$- Median$\rvert$**
7	3	2
4	6	5
10	0	1
9	1	0
15	5	6
12	2	3
7	3	2
9	1	0
9	1	0
18	8	9
Total =100	**30**	**28**

Mean deviation from mean $= \frac{\sum |D|}{n} = \frac{30}{10} = 3$

Co-efficient mean deviation about mean $= \frac{\text{Mean deviation about mean}}{\bar{x}} = \frac{3}{10} = 0.3$

Mean deviation about median $= \frac{\sum |Dm|}{n}$

$= \frac{28}{10} = 2.8$

Co- efficient mean deviation about median $= \frac{\text{Mean deviation about median}}{\text{median}}$

$= \frac{2.8}{9} = 0.311$

Ch6_11th Stats.indd 162 2/3/2019 11:01:55 AM